

# \* Analysis of Variance (ANOVA)

## Hypothesis Testing in ANOVA

(Partitioning of Variance in the Anova)

Null Hypothesis  $H_0: \mu_1 = \mu_2 = \mu_3 \dots \mu_k$

Alternate Hypothesis  $H_1$ : Alternate of one of the sample mean is not equal.

$\mu_1 \neq \mu_2 \neq \mu_3 \dots \mu_k$  /  $\chi \rightarrow$  we can't use this.

## Test Statistics:

$$F = \frac{\text{Variance between sample}}{\text{Variance within sample}}$$

| variance b/w sample    |                 |                            |                 |
|------------------------|-----------------|----------------------------|-----------------|
|                        | $x_1$           | $x_2$                      | $x_3$           |
| Variance within sample | 1               | 6                          | 5               |
|                        | 2               | 7                          | 6               |
|                        | 4               | 3                          | 3               |
|                        | 5               | 2                          | 2               |
|                        | 3               | 1                          | 4               |
|                        | $\bar{x}_1 = 3$ | $\bar{x}_2 = \frac{19}{5}$ | $\bar{x}_3 = 4$ |

$$H_0: \bar{x}_1 = \bar{x}_2 = \bar{x}_3$$

$H_1$ : Atleast one sample mean is not equal.



One Way ANOVA: One factor with atleast 2 levels.  
levels are independent.

- ① Doctor want to test a new medication which reduces headache. They split the participant into 3 conditions [15 mg, 30 mg, 45 mg]. Later on the doctor ask the patient to rate the headache b/w [1-10]. Are there any differences b/w the 3 conditions,  $\alpha = 0.05$ ?

| 15 mg | 30 mg | 45 mg |
|-------|-------|-------|
| 9     | 7     | 4     |
| 8     | 6     | 3     |
| 7     | 6     | 2     |
| 8     | 7     | 3     |
| 8     | 8     | 4     |
| 9     | 7     | 3     |
| 8     | 6     | 2     |

- i. Define Null and Alternate hypothesis.

$$H_0: \mu_{15} = \mu_{30} = \mu_{45}$$

$H_1$ : not all  $\mu$  are equal.

- ii. Significance  $\alpha = 0.05$   $CF = 0.95$

- iii. Calculate Degree of freedom

3 rows  $\rightarrow N = 21$ ,  $a = 3$ ,  $n = 7$   
3 columns

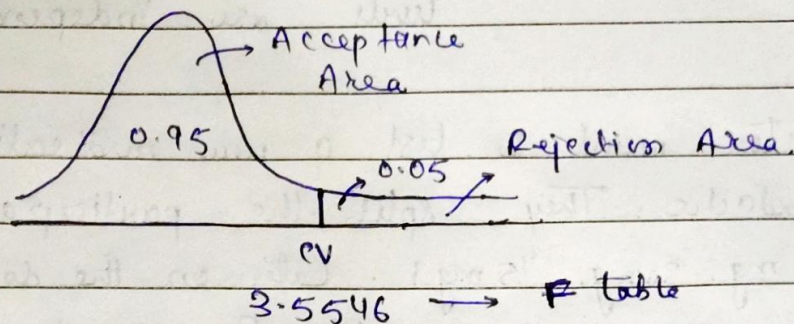
$$df_{\text{between}} = a - 1 = 3 - 1 = 2 \quad \left. \begin{array}{l} df_1 \quad df_2 \\ (2, 18) \end{array} \right\}$$

$$df_{\text{within}} = N - a = 21 - 3 = 18$$

$$df_{\text{total}} = N - 1 = 21 - 1 = 20 \quad \left. \begin{array}{l} \text{f table} \\ \alpha = 0.05 \end{array} \right\} \rightarrow \text{critical value}$$



## 10. Decision Boundary:



## Decision Rule

If  $f$  is greater than 3.5546, reject the  $H_0$ .

## v. Calculate F test Statistic

|         | SS     | df | MS    | F | SS → Sum of square  |
|---------|--------|----|-------|---|---------------------|
| Between | 98.67  | 2  | 49.34 |   |                     |
| Within  | 10.29  | 18 | 0.54  |   | MS → Mean of square |
| Total   | 108.96 | 20 |       |   |                     |

$$i. \text{ SS between} = \frac{\sum (\sum a_i)^2}{n} - \frac{T^2}{N}$$

$$15 \text{ mg: } 9+8+7+8+8+9+8 = 57$$

$$30 \text{ mg: } 7+6+6+7+8+7+6 = 47$$

$$45 \text{ mg: } 4+3+2+3+4+3+2 = 21$$

$$= \frac{57^2 + 47^2 + 21^2}{7} - \frac{[57^2 + 47^2 + 21^2]}{21}$$

$$= \underline{98.67}$$



ii. SS within:  $\sum y^2 - \frac{\sum (a_i)^2}{n}$

$$\sum y^2 = 9^2 + 8^2 + 7^2 + 8^2 + 8^2 + 9^2 + 8^2 + \dots$$

$$= 853$$

$$= 853 - \frac{[57^2 + 47^2 + 21^2]}{7}$$

$$= \boxed{10.29}$$

F test =  $\frac{\text{MS Between}}{\text{MS within}}$

$$= \frac{49.84}{0.54} = 86.56$$

If  $f$  is greater than 3.5546, Reject the  $H_0$ .

$$86.56 > 3.5546 \quad \text{Reject the } H_0.$$